Supplementary material to "Information Budgets of Trajectory Health Indicators for Remaining Useful Life Prediction"

S1. Initial benchmark protocol (P1)

Abbreviations are those of the main text: health indicator (HI), remaining useful life (RUL), root mean squared error (RMSE), mean absolute error (MAE), continuous ranked probability score (CRPS), physics-guided information bottleneck (PGIB), variational information bottleneck (VIB), β-variational autoencoder (β-VAE), first-principal-component HI (PCA-HI), gated recurrent unit (GRU), Center for Advanced Life Cycle Engineering (CALCE), and the FEMTO-ST/PRONOSTIA bearings (FEMTO); the age law is the lifetime distribution conditioned on survival to the current age.

The first benchmark used a simpler protocol (P1). Every HI was calibrated on in-sample training paths, the plain first-hitting-time law (B.2) was used without age fusion, and the predictive grid extended to three times the held-out unit's own life, which leaks the length of the test unit into the support of the predictive law (removed in P2). The pilot on NASA B0005, the results below and the observation that the age law is competitive motivated P2. The P1 results are reported for completeness and are not used for the conclusions of the main text.

Pilot run. A pilot on the NASA B0005 fold, with a GRU-only encoder and $\beta \in \{0.003, 0.03, 0.3\}$, gave a PGIB RMSE of 56.3 cycles and zero interval coverage, with the selected β at the edge of the grid. The selection grid was therefore extended and the feed-forward encoder added as a candidate. Both changes are chosen per fold by inner validation, but the decision to introduce them was made after seeing a held-out result for one fold. No baseline was modified, and no change was made after the full runs reported below; the bearing experiments were run once with the final protocol.

Results under P1. The outcome depends on the data set (Tables 1 and 2, Figure 1), and the age law is needed to read it. On NASA the age-law RMSE is 39.8 cycles; PCA-HI (17.8), the Geo-opt HI (21.9), the β-VAE (30.8) and VIB (33.3) beat it, whereas the PGIB (47.6) does not, and its intervals cover only 42% of the true RULs. On CALCE the age-law RMSE is 49.3; only the PGIB (33.0) beats it clearly, while PCA-HI, the Geo-opt HI and the β-VAE (117 to 164) are considerably worse than the age law. With the two 0.5C-discharge cells added, the PGIB keeps the lowest RMSE (50.4) but its margin over the age law (54.7) is only 8%. On bearings the age-law RMSE (602 and 664 snapshots for conditions 1 and 2) is comparable to the standard deviation of the pooled RUL, and most HIs do no better: the PGIB has the lowest RMSE in condition 1 (524.5, 13% below the age law) but gives almost no benefit in condition 2 (621.8 against 664). Two long-lived units dominate the pooled error in condition 2.

The ablations (in the supplied result files) support the bottleneck on all five sets (for example RMSE 54.4 $\rightarrow$ 47.6 on NASA and 42.3 $\rightarrow$ 33.0 on CALCE) and the epistemic drift variance for coverage (for example 14 $\rightarrow$ 42% on NASA). The physics prior helps on CALCE and FEMTO condition 1 but not on NASA or FEMTO condition 2. With only three to six training units per fold, the validation signal that selects the encoder is itself noisy; on NASA it chose the strongest compression ($\beta = 30$), at the edge of the grid, in all eight selections.

Table 1: Protocol P1, battery benchmarks, leave-one-unit-out (mean ± standard deviation over seeds). RMSE, MAE and CRPS in cycles.

NASA PCoE (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	39.8	34.7	–	–	–
PCA-HI	0.27	0.85	0.77	17.8	14.0	97	2.52	11.1
Geo-opt HI	0.21	0.93	0.95	21.9	17.0	90	2.43	12.8
β-VAE-HI	0.30±0.01	0.71±0.00	0.95±0.01	30.8±0.8	26.0±0.5	95±1	2.06±0.08	17.6±0.5
VIB-HI (no physics prior)	1.00	1.00±0.00	0.67±0.03	33.3±2.8	27.7±2.1	88±4	3.47±0.10	20.5±2.3
PGIB-HI	1.00	1.00±0.00	0.67±0.01	47.6±2.1	42.3±1.9	42±3	2.39±0.15	34.3±1.5
CALCE CS2 (4 cells, 1C/1C) (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	49.3	43.8	–	–	–
PCA-HI	0.05	0.72	0.83	130.5	110.4	92	2.74	82.3
Geo-opt HI	0.39	0.95	0.98	117.2	86.8	90	3.20	69.2
β-VAE-HI	0.02±0.01	0.72±0.01	0.80±0.01	163.7±0.7	148.0±0.9	83±1	1.87±0.02	111.1±0.7
VIB-HI (no physics prior)	1.00	1.00±0.00	0.92±0.01	53.2±0.9	47.2±1.4	71±4	1.24±0.13	38.4±0.7
PGIB-HI	1.00	1.00±0.00	0.95±0.01	33.0±4.9	29.9±3.9	55±8	0.34±0.02	22.6±3.1
CALCE CS2 (6 cells, incl. 0.5C-discharge CS2_33/34) (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	54.7	50.0	–	–	–
PCA-HI	0.13	0.28	0.21	191.8	172.0	50	0.99	149.6
Geo-opt HI	0.39	0.95	0.98	106.9	82.7	95	3.11	63.7
β-VAE-HI	0.05±0.02	0.52±0.10	0.33±0.04	201.6±5.3	179.9±5.0	48±0	1.46±0.18	156.1±3.1
VIB-HI (no physics prior)	1.00	1.00±0.00	0.87±0.02	69.7±6.0	54.4±8.8	90±5	2.24±0.23	44.9±4.8
PGIB-HI	1.00	1.00±0.00	0.91±0.01	50.4±4.4	44.7±5.1	69±5	0.58±0.02	33.4±4.2

Table 2: Protocol P1, FEMTO bearing benchmarks. RMSE, MAE and CRPS in 10 s snapshots.

FEMTO bearings, condition 1 (7 units) (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	602.5	515.4	–	–	–
PCA-HI	0.06	0.56	0.70	784.5	663.2	93	2.80	485.7
Geo-opt HI	0.04	0.29	0.39	770.2	623.3	91	3.34	459.1
β-VAE-HI	0.07±0.02	0.58±0.01	0.73±0.01	779.8±51.1	664.4±35.6	91±3	2.69±0.13	492.4±20.6
VIB-HI (no physics prior)	1.00	1.00±0.00	0.71±0.02	570.1±56.0	471.0±50.7	78±5	3.04±0.21	365.6±44.1
PGIB-HI	1.00	1.00±0.00	0.71±0.02	524.5±45.3	423.6±37.3	75±5	2.65±0.10	336.5±31.8
FEMTO bearings, condition 2 (7 units) (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	664.4	584.6	–	–	–
PCA-HI	0.04	0.43	0.17	613.9	487.6	54	2.25	433.9
Geo-opt HI	0.06	0.37	0.46	559.2	457.1	73	2.33	356.5
β-VAE-HI	0.06±0.01	0.46±0.05	0.42±0.16	539.8±31.7	439.6±24.6	70±2	2.01±0.06	348.6±26.1
VIB-HI (no physics prior)	1.00	0.99±0.00	0.39±0.02	616.8±1.5	493.8±1.5	41±1	2.62±0.23	451.1±3.2
PGIB-HI	1.00	0.99±0.00	0.46±0.02	621.8±4.3	498.2±8.1	36±2	2.25±0.01	464.3±7.9

Table 3: Protocol P1, information certified by the achieved error (Theorem 1(b)), in nats; < 0: vacuous. $h(T)$: entropy of the pooled held-out RUL.

Dataset	$h(T)$	PCA	Geo-opt.	β-VAE	VIB	PGIB
NASA	4.36	0.06	< 0	< 0	< 0	< 0
CALCE (4)	5.67	< 0	< 0	< 0	0.27	0.76
CALCE (6)	5.66	< 0	< 0	< 0	< 0	0.32
FEMTO c. 1	7.24	< 0	< 0	< 0	< 0	< 0
FEMTO c. 2	7.04	< 0	< 0	< 0	< 0	< 0

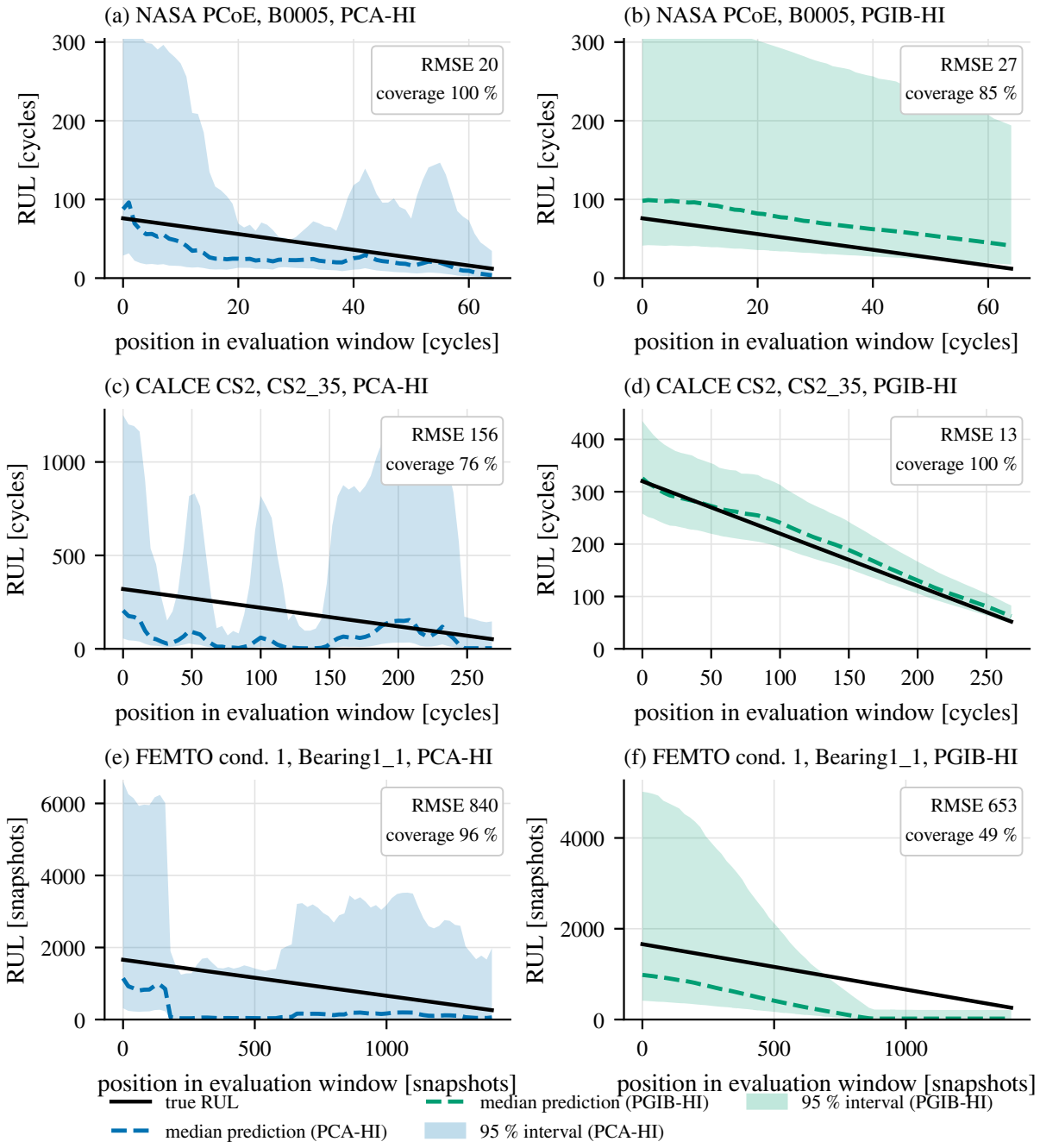


Figure 1: Example RUL predictions with 95% intervals on one held-out unit per data set (NASA B0005, CALCE CS2_35, FEMTO Bearing1_1), for PCA-HI (left) and PGIB (right); each panel gives the RMSE and the coverage on that unit.

Geometric scores versus error. Across the five HIs of each data set, the rank correlation between the geometric score Mon + Tre + Pro and the RMSE is +0.90 on NASA (a higher geometric score goes with a larger error), −1.00 on CALCE, −0.90 on six-cell CALCE, −0.60 on FEMTO condition 1 and +0.60 on condition 2. The geometric criteria therefore did not rank the HIs consistently by prognostic error. This is an association over five methods per data set and not evidence for the mechanism of Theorem 3: the VIB and PGIB HIs have Mon ≈ Tre ≈ 1 because their velocity is constrained to be positive, not because of a squeeze.